

Evidence-Gated Adaptive Ingestion

FRONTIER INGEST | 1/3

One production pipeline, two workload lanes, and a hard rule: no structured field without source evidence.

CATALOG LANE

Bulk pages, checkpointed batches: 1,000 startups, products and papers.

FRESHNESS LANE

Strict timestamp proof: all qualifying jobs and news from the last 24 hours.

1 Discover + enqueue

APIs, feeds, stable dedupe keys

2 Fetch once

Rate limits, retries, block detection

3 Raw evidence

SHA-256 content store

4 Extract

JSON, feeds, JSON-LD, HTML

8 Ack + export

Idempotent upsert, metrics, six tabs

7 Validate + link

Freshness, schema, entities

6 Targeted LLM

Small fragments, typed output

5 Evidence gate

Only unresolved fields continue

Deterministic first

Structured source fields consume zero LLM tokens.

Field-level lineage

Every record carries source URL, collected time and content hash.

Safe abstention

Uncertain freshness or entity matches are rejected or flagged for review.

Source plan and token economics

Use the cheapest trustworthy representation first; reserve language models for genuine ambiguity.

Monitored source inventory

Dataset	Primary sources	Captured evidence	Access and validation
Startups	Y Combinator AI company directory	Canonical + raw name, team size, website, description, batch and status	Paginated public directory; one canonical company URL per record.
AI products	Hugging Face Spaces public API	Product title, publisher, description, tags, SDK, likes and timestamps	Cursor pagination; one genuine hosted AI application per record; pricing abstains without evidence.
Research papers	Hugging Face daily papers + arXiv + GitHub GraphQL	Title, authors, abstract, paper URL, repository URL, stars and star timestamp	Code-linked papers only; GitHub enrichment batched up to 40 repositories per call.
AI jobs	Arbeitsnow, Remote OK, Jobicy, Himalayas, Remote	Company, title, published time, remote flag, location and full description	Five official/public APIs; provable age <= 24h; source-specific attribution honored.
AI news	OpenAI, DeepMind, Hugging Face, MIT News AI, TechCrunch AI	Title, publisher, authors, published time and full article text	RSS/Atom discovery followed by one compliant article fetch; age <= 24h.

Token budgeter: selection, not summarization

- **1. Parse locally**
APIs, feeds, metadata and JSON-LD fill known fields at zero token cost.
- **2. Detect the gap**
The evidence gate names only missing or conflicting fields.
- **3. Select fragments**
Headings and nearby sentences are scored; whole pages are never sent.
- **4. Enforce hard limits**
Per-record, per-source and per-run budgets stop token drift.
- **5. Validate or abstain**
Typed output must cite supplied evidence; retry once, then flag review.

Access policy and anti-bot boundary

- Priority: official API -> RSS/Atom -> sitemap -> permitted HTTP -> allowlisted browser rendering only when authorized.
- Per-host concurrency, minimum delays, exponential retry and jitter prevent accidental overload.
- Robots rules, terms, cache headers and attribution are source configuration, not ad hoc code.
- The executable Playwright adapter accepts authorized session state; challenge signatures stop it immediately.
- Raw-response hashes deduplicate downloads and allow parser replays without touching the source again.

Representative endpoints: ycombinator.com/companies/industry/ai | huggingface.co/api/spaces | huggingface.co/api/daily_papers
Freshness inputs: arbeitsnow.com/api/job-board-api | remoteok.com/api | jobicy.com/api/v2/remote-jobs | remote.com/api/remote-jobs

500,000-record production design and verification

At-least-once processing becomes safe through stable keys, leases, immutable evidence and idempotent upserts.

Scale-out topology



Capacity

- Pages stream; no full-dataset memory load.
- 500k hot path: 2,500 batches; 0.002 MiB traced peak.
- Autoscale on queue lag and oldest-item age.

Correctness

- Stable key + unique constraint = idempotency.
- SKIP LOCKED leases + queue-order-independent reconciliation.
- Canonical output preserves the raw source name.

Operations

- Persist source yield, queue, HTTP and LLM metrics.
- 413 shrinks payload; 429 backs off then falls back.
- Dead letters retain raw evidence and error.

Storage choice: PostgreSQL is the ACID source of truth with JSONB and pgvector candidate search; Neo4j is an optional, rebuildable projection for multi-hop traversal.

Submission quality gate - final verified run on 12 Sep 2026

Dataset	Rows	Unique keys	Valid source URLs	Evidence complete	Result
AI startups	1,000	1,000	1,000	1,000	PASS
Genuine AI products	1,000	1,000	1,000	1,000	PASS
Research papers + code	1,000	1,000	1,000	1,000	PASS
AI jobs, verified <= 24h	67	67	67	67	PASS
AI news, verified <= 24h	11	11	11	11	PASS

Fail-closed rules enforced: direct GitHub metrics must be <= 24h old; unverifiable signal dates are excluded.